

Gregory Chisholm

AI Agent & Workflow Automation Engineer / Cloud Infrastructure

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SUMMARY

Engineer with 8+ years shipping production web applications, now building AI agents and workflow automation on AWS. I ship complete products rather than demos: a Shopify app with paying merchants, four live automation systems, and a browser-based voice agent. My focus is the part that decides whether an agent is a demo or a product, meaning least-privilege access, hard spend ceilings, audit trails, and defenses against prompt injection. AWS Solutions Architect Associate certified.

TECHNICAL SKILLS

AWS	Lambda, API Gateway, S3, CloudFront, DynamoDB, IAM, SSM Parameter Store, CloudWatch, Route 53
AI / Agents	OpenAI and Anthropic APIs, Realtime voice (WebRTC), function/tool calling, structured outputs, retrieval and embeddings, prompt injection defense, MCP
Security	GitHub OIDC federation, least-privilege IAM, secrets management, CSP and security headers, OAuth flows, GDPR compliance
Automation	Document extraction pipelines, web scraping and change detection, schema-enforced LLM output, deterministic validation layers, cost modeling
Languages	JavaScript (primary), Node.js, HTML/CSS, SQL, Python, TypeScript
Tooling	Git, GitHub Actions, Terraform, Docker, Linux/bash, Vue, Next.js, Jira

EXPERIENCE

Founder & Sole Engineer, Chisholm Digital Assets LLC

Feb 2025 - Jan 2026

DripCheck, a virtual try-on app published on the Shopify App Store

- Shipped a serverless AWS application end to end (Lambda, API Gateway, S3, DynamoDB) integrating an asynchronous image-generation API, with per-store usage metering and quota enforcement.
- Cut the AWS bill by over 99% by re-architecting from an always-on EC2 instance to Lambda, moving cost from wall-clock hours to per-invocation and turning per-transaction margin positive.
- Passed Shopify App Store review: OAuth install flow, mandatory GDPR compliance webhooks, Billing API integration, and embedded-app requirements.
- Sole engineer across backend, frontend, infrastructure, billing, and support. Wound the product down deliberately after foundational image models commoditized the core feature: deprecation notice, billing terminated, infrastructure decommissioned.

Developer 2, iHerb / Vitacost

Nov 2017 - Aug 2026

Web engineering on a health and wellness e-commerce site, 1.3M+ sessions per month

- Took over a promo-deployment tool and rewrote its output so it required no hand-editing, cutting each promotional placement from 5 minutes to about 60 seconds across hundreds of promos per week.
- Rebuilt the site navigation and validated it with a 50/50 A/B test, measuring roughly a 10% reduction in desktop bounce rate; shipped permanently.
- Owned privacy-policy implementation with OneTrust alongside the privacy team, keeping site functionality compliant with US and EU regulations including GDPR.
- Led frontend work on the Vitacost to Shopify migration, rebuilding pages and the reusable component library the team assembled new pages from.
- Remediated accessibility on high-traffic COVID-19 information pages using Axe DevTools.
- Promoted from Developer 1; scope grew from individual campaign pages to owning business initiatives and the unfamiliar problems the team had not tackled before.

SELECTED PROJECTS

Voice Agent / OpenAI Realtime, WebRTC, Lambda, DynamoDB, S3

- Browser-based voice agent that answers questions about my work from a hand-authored knowledge base. No API credential ever reaches the browser; the backend brokers the connection and stays out of the audio path. Architecture and threat model published at gregorychisholm.com/talk/architecture.

- Retrieval-grounded: every factual answer comes from an embedded, version-controlled corpus with per-skill honesty levels, and the agent abstains rather than guessing when there is no source. Retrieval quality is measured against a 49-case eval suite.
- Enforced in code outside the model's reach: kill switch, per-visitor and global rate limits, pre-charged daily spend ceiling, output-token cap, fail-closed limiters.

Document Extraction Pipelines / Vision LLMs, Next.js, Vercel

- Two live systems converting receipts and legal contracts into validated structured data. Every contract risk flag must cite a verbatim clause, and code then verifies that quote exists in the source document, giving effectively free hallucination detection.
- Benchmarked five vision models against hand-verified and attorney-annotated ground truth (CUAD). The cheapest model was also the most accurate, at roughly 19x lower cost than the frontier model that scored worse. Disqualified one model for a decimal-place error despite a good average.

Press Release Monitor / Firecrawl, LLM summarization, tiered cost architecture

- Monitors corporate newsrooms for new releases and emails alerts. Probing real sites first showed most block ordinary scrapers, including one returning HTTP 200 with an empty JavaScript shell, which would make a naive monitor silently blind.
- Three-tier pipeline (free RSS, then scrape-and-diff, then extraction only on real change) brought a projected \$749/month API cost down to \$99/month at 119,000 checks per month.

gregorychisholm.com / S3, CloudFront, Lambda, GitHub Actions

- Replaced a 344-day-old static AWS access key in CI with GitHub OIDC federation, scoping the trust policy to one repository and branch and reducing the deploy role to four actions on three named resources.
- Took the site from an F to an A on securityheaders.com with a hand-tuned CSP, HSTS, and Permissions-Policy. Wrote it up publicly, including the remaining weakness in my own policy.

Cloudy Calc / Chrome extension, ~336 users, 4.9/5

- Keyboard-driven calculator published on the Chrome Web Store and actively maintained. Collects no user data by design.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (valid through 2028)
- HashiCorp Certified: Terraform Associate (003)
- Google Cloud Associate Cloud Engineer
- AWS Certified Cloud Practitioner
- In progress: Certified Kubernetes Administrator (CKA), AWS Certified CloudOps Engineer - Associate

EDUCATION

B.S. Psychology, University of Central Florida | CS50x Miami, full-time in-person software development program